

BIS TRANSPORT ENGINEERING DEPARTMENT (TED)

RAG / PDF Ingestion Knowledge Base — Indian Standards, Technical Committees and Domain Guidance

Purpose: This document is structured as a retrieval-oriented knowledge base for an AI assistant. It summarizes the BIS Transport Engineering Department (TED), its technical committee structure, major subject areas, and a curated set of important standards and standard families. It is not a replacement for the latest official Indian Standard text.

Primary source basis: BIS Transport Engineering Department publication on IC Engine Components / TED, BIS Connect standard records, and BIS standard-detail records accessed in September 2026.

1. Department Identity and Scope

Department: Transport Engineering Department (TED)

Division Council: Transport Engineering Division Council (TEDC)

Official scope: Standardization in transport engineering including air, water, road and rail transport; diesel engines for stationary application; freight containers; transport packaging; road and transport safety; and related areas.

BIS describes TED as one of its technical departments. TED develops Indian Standards through sectional technical committees covering automotive engineering, tyres, electrical equipment, air and space vehicles, bicycles, shipbuilding, marine engineering, transport packaging, alternative-energy vehicles, electric and hybrid vehicles, intelligent transport systems, passive safety, in-service inspection, unmanned aerial systems, freight containers, suspension systems and road safety.

2. TED Standards Portfolio — Reference Snapshot

Aspect	Number of standards (as of 15 Sep 2023)
Product	461
Code of Practice	105
Methods of Test	211
Terminology	65
Dimensions	72
System Standard	7
Safety Standard	10
Others	253
Service Specification	3
Process Specification	1
Unclassified	73
Total	1709

These figures are a historical portfolio snapshot from the BIS TED publication, not a current count. Individual standards must be checked against the current BIS standard record for status, amendments and review/reaffirmation.

3. TED Sectional Technical Committees

TED 02 — Automotive Primemovers, Transmission Systems and Internal Combustion Engine

Scope: Automotive engines and transmissions; components; stationary internal-combustion engines for agriculture.

TED 04 — Automotive Braking Systems, Vehicle Testing, Steering and Performance Evaluation

Scope: Braking and steering systems/components; whole-vehicle testing; performance evaluation and roadworthiness.

TED 06 — Automotive Body, Chassis and Accessories

Scope: Chassis assembly; aerodynamics; ergonomics; vehicle body; body fittings; seats; accessories; garage equipment.

TED 07 — Automotive Tyres, Tubes and Rims

Scope: Automotive tyres, tubes, rims and related components.

TED 11 — Automotive Electrical Equipment and Instruments

Scope: Electrical and optical equipment, instruments and related components, including automotive electronics.

TED 14 — Air and Space Vehicles

Scope: Aircraft/space-vehicle systems, accessories, fittings, test methods and performance; aircraft electrical equipment; air-cargo and specialized ground-handling equipment.

TED 16 — Bicycles

Scope: Bicycles, electric bicycles, parts, accessories and terminology.

TED 17 — Shipbuilding

Scope: Ships and systems/accessories/fittings, performance, testing, terminology, codes of practice, marine environment, ship management and quality assurance.

TED 18 — Inland, Harbour Crafts and Fishing Vessels

Scope: Hull/deck fittings and constructional practices for inland vessels, dredgers, tugs, barges, launches and fishing vessels.

TED 19 — Marine Engineering and Safety Aids

Scope: Marine propulsion/auxiliary machinery and fittings; testing; safety and lifesaving equipment; fire protection/detection; navigation and signalling equipment.

TED 22 — Transport Tractors, Trailers and Industrial Trucks

Scope: Transport/industrial tractors, trailers including container trailers, powered industrial trucks, attachments, components, testing and operating/maintenance safety.

TED 24 — Transport Packages, Packaging Codes and Pallets

Scope: Transport packages, packaging tests, test intensities, packaging codes and pallet performance requirements/methods of test.

TED 26 — Automotive Vehicles Running on Non-Conventional Energy Sources

Scope: Fuel-system components and installation requirements for CNG, LPG, biogas, hydrogen/methane blends, biodiesel/diesel blends, alcohol-petrol blends, hydrogen and fuel-cell applications.

TED 27 — Electric and Hybrid Vehicles

Scope: Electric and hybrid vehicles and their components.

TED 28 — Intelligent Transport Systems

Scope: Information, communication and control systems for urban/rural surface transport, intermodal/multimodal transport, traveller information, traffic management, public/commercial transport and emergency services.

TED 29 — Passive Safety Crash Protection Systems

Scope: Passive safety and crash-protection systems and impact-test procedures.

TED 31 — In-service Vehicle Inspection, Certification and Garage Equipment

Scope: Vehicle inspection/certification, visual inspection for periodic technical inspection and associated garage equipment.

TED 32 — Unmanned Aerial Vehicles

Scope: Unmanned aircraft systems including classification, design, manufacture, operation/maintenance, insurance and safety management for civil, medical, military, agricultural and other uses.

TED 33 — Freight Containers

Scope: General and specific-purpose freight containers; test methods; performance; identification/marketing/communication; safety and handling/securing equipment.

TED 34 — Springs and Suspension Systems

Scope: Shock absorbers and springs, including automotive engine rocker-arm springs and suspension systems.

TED 35 — Road Safety

Scope: Road safety, associated equipment/accessories and road-safety management systems.

4. High-Value Automotive Standards and Standard Families

IS Number	Title	Why it matters / classification
IS 12222:2011	Automotive vehicles — Turning circle and maneuverability requirements (First Revision)	Vehicle (First Revision) / steering performance. BIS Connect record: rev
IS 11825:2023	Automotive Vehicles — Method of Weighment (First Revision)	Vehicle weighing/test method. TED 04; methods of test.
IS 11140:2024	Automotive Vehicles — Footrests for Two-Wheeled Motor Vehicle Two-Wheelers (First Revision)	Two-Wheelers (First Revision) TED 06; product specification.
IS 13154:2015	Automotive vehicles — Tyres for agricultural vehicles and their trailers (First Revision)	Agricultural Specification (First Revision) TED 07.
IS 1062:2014	Road vehicles — Spark-plugs — Test methods and requirements (First Revision)	Spark-plugs (First Revision) methods and requirements. TED 11; reaffirmed March 20
IS 13599:2014	Automotive vehicles — Instrument systems — Temperature gauges and Electric and Electronic Instrument Specifications (First Revision)	Gauges and Electric and Electronic Instrument Specifications (First Revision)
IS 19281 (Part 1):2025	Road vehicles — Compressed gaseous hydrogen (CGH2) and hydrogen/natural-gas blends systems (Part 1) Safety requirements	Hydrogen/natural-gas blends systems (Part 1) Safety requirements
IS/ISO 12619-14:2017	Road vehicles — CGH2 and hydrogen-natural-gas blends fuel systems (Part 14) Fuel system component	Hydrogen/natural-gas blends fuel systems (Part 14) Fuel system component. TED 26; product specifica

5. Internal-Combustion Engine and Generator Standards

Standard / family	Subject	Knowledge-base description
IS 10000 series	Methods of tests for internal-combustion engines	Includes terminology, reference conditions, measurement accuracy, test pre
IS/ISO 8528 series	Reciprocating internal-combustion engine driven AC generator sets	Generator sets application/rating/performance, control gear and switchgear, genera
IS 10001:1981	Performance requirements for constant-speed compression ignition (diesel) engines for small constant speed diesel engines.	Performance (diesel) engines for small constant speed diesel engines.
IS 10002:1981	Performance requirements for constant-speed compression ignition (diesel) engines for large constant speed diesel engines.	Performance (diesel) engines for large constant speed diesel engines.
IS 11170:1985	Performance requirements for constant-speed compression ignition (diesel) engines for agricultural and power up to 20 kW	Agricultural (diesel) engines for agricultural and power up to 20 kW
IS 14599:1999	Automotive vehicles — Performance requirements (measurement of power, S.F.O capacity) for compression ignition engine	Engine performance (S.F.O capacity) for compression ignition engine
IS 14553:2008	Automotive vehicles — Apparatus for measurement of opacity (smoke) of exhaust gas from engines equipped with compression-ignition	Opacity (smoke) of exhaust gas from engines equipped with compression-ignition

6. Frequently Relevant TED Knowledge for an AI Assistant

What does TED mean in BIS?

TED means Transport Engineering Department. It is the BIS technical department responsible for standardization in transport engineering and associated subjects.

What does TED cover?

Air, water, road and rail transport; stationary diesel engines; freight containers; transport packaging; road and transport safety; and related transport-engineering subjects.

What is TEDC?

TEDC is the Transport Engineering Division Council associated with the Transport Engineering Department.

Which committee covers automotive braking and roadworthiness?

TED 04 — Automotive Braking Systems, Vehicle Testing, Steering and Performance Evaluation.

Which committee covers tyres, tubes and rims?

TED 07.

Which committee covers automotive electrical equipment and instruments?

TED 11.

Which committee covers bicycles and electric bicycles?

TED 16.

Which committee covers ships and ship systems?

TED 17; marine/inland craft subjects are also covered by TED 18 and TED 19 according to their respective scopes.

Which committee covers alternative-fuel automotive vehicles?

TED 26.

Which committee covers electric and hybrid vehicles?

TED 27.

Which committee covers intelligent transport systems?

TED 28.

Which committee covers crash/passive safety?

TED 29.

Which committee covers in-service vehicle inspection and certification?

TED 31.

Which committee covers unmanned aircraft systems?

TED 32.

Which committee covers freight containers?

TED 33.

Which committee covers springs and suspension systems?

TED 34.

Which committee covers road safety?

TED 35.

7. Important Retrieval Rules for the BIS AI Assistant

- Always distinguish the **standard number** from the **product/domain**. A user may describe a product without knowing its IS number.
- Do not infer that every TED standard is mandatory certification. A standard can be voluntary, referenced by regulation, used for testing, or otherwise applicable. Check the current BIS record and applicable regulation/scheme before stating that certification is compulsory.
- Always check the **current BIS standard record** for status, amendments, revisions, reaffirmation/review information, certification details and referenced standards.
- Where a standard has parts, preserve the complete designation, for example **IS/ISO 8528-5** or **IS 10000 (Part 8)**.
- Separate **product specification**, **method of test**, **code of practice**, **safety standard** and **terminology**. They answer different user questions.
- For compliance questions, identify the product, intended use, vehicle/category, fuel/energy type, and whether the question concerns design, testing, manufacturing, certification or in-service inspection.
- Do not provide clause-level requirements from memory. For exact limits, dimensions, test conditions, sampling plans or acceptance criteria, retrieve the current standard text or official BIS record.

8. Source and Currency Note

The department-level structure and committee scopes in this knowledge base are based primarily on the official BIS TED publication. Individual standard metadata is based on BIS Connect / BIS Standards records retrieved during preparation. Because Indian Standards can be revised, amended, reaffirmed, withdrawn or replaced, the AI assistant should treat the standard number/title as a retrieval key and verify current status before giving regulatory or compliance advice.

Official BIS knowledge locations: BIS website, BIS Connect / Know Your Standards, and BIS Standards standard-detail records.

End of TED Knowledge Base